VII SEMESTER (OPEN ELECTIVE IV)	L	T	P	C
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20EE7O02 - INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLER				

COURSE OUTCOMES:

After successful completion of this course, students should be able to:

CO1: Illustrate I/O modules of PLC systems and ladder diagrams.

CO2: Demonstrate various types registers and programming instructions.

CO3: Examine various types of PLC functions and its applications.

CO4: Assess different data handling functions and its applications.

CO5: Describe the analog operations and PID modules

UNIT-I: INTRODUCTION TO PLC SYSTEMS

I/O modules and interfacing - CPU processor - programming Equipment - programming formats - construction of PLC ladder diagrams - Devices connected to I/O Modules. Digital logic gates - programming in the Boolean algebra system - conversion examples Ladder Diagrams for process control: Ladder diagrams & sequence listings - ladder diagram construction and flowchart for spray process system

UNIT-II: PLC PROGRAMMING & REGISTERS

PLC Programming: Input instructions - outputs - operational procedures - programming examples using contacts and coils. Drill press operation.

PLC Registers: Characteristics of Registers - module addressing - holding registers - Input Registers - Output Registers.

UNIT-III: PLC FUNCTIONS

Timer functions & Industrial applications - counters - counter function industrial applications - Arithmetic functions - Number comparison functions - number conversion functions

UNIT-IV: DATA HANDLING FUNCTIONS

SKIP - Master control Relay - Jump - Move - FIFO - FAL - ONS – CLR & Sweep functions and their applications. Bit Pattern and changing a bit shift register – sequence functions and applications - controlling of two-axis & three axis Robots with PLC - Matrix functions

UNIT-V: ANALOG PLC OPERATION

Analog modules & systems - Analog signal processing - Multi bit Data Processing - Analog output Application Examples - PID principles - position indicator with PID control - PID Modules - PID tuning - PID functions.

Text Books:

1. Programmable Logic Controllers- Principles and Applications by John W. Webb & Ronald A. Reiss - Fifth Edition – PHI.
2. Programmable Logic Controllers- Programming Method and Applications –JR.Hackworth & nF.D Hackworth Jr. –Pearson - 2004

Reference Books:

1. Introduction to Programmable Logic Controllers- Gary A. Dunning - 3rd edition – Cengage Learning - 2005.
2. Programmable Logic Controllers –W.Bolton - 5th Edition - Elsevier publisher - 2009.